

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)**REVIEW TEST****06-09-2020****JEE(Main + Advanced) : NURTURE COURSE [PHASE : PHASE-TNPS, TOPS & TRPS]****PAPER-1****Time : 3 Hours****Maximum Marks : 180****READ THE INSTRUCTIONS CAREFULLY****GENERAL :**

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR**Please see the last page of this booklet for rest of the instructions**

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

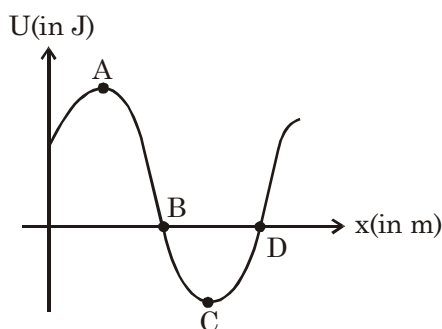
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. In the given figure the variation of potential energy of a particle of mass 'm' is represented w.r.t. its x-coordinate. The particle can move under the effect of this conservative force along the x-axis. Select the **CORRECT** alternatives :



- (A) At point B the acceleration of particle is along the positive x-axis.
- (B) At point C the acceleration of particle is along the positive x-axis.
- (C) At point A and C particle is in unstable and stable equilibrium respectively.
- (D) At point B and D particle is in stable and unstable equilibrium respectively.

Space for Rough Work

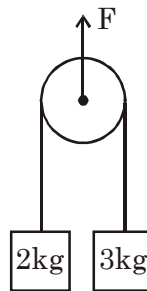
2. A force $\vec{F} = 3\hat{i} + 2\hat{j} + C\hat{k}$ N. causes a displacement $\vec{r} = C\hat{i} + 4\hat{j} + C\hat{k}$ m. The work done is 36 J. Find the value of C :-
 (A) 4 (B) 5 (C) -7 (D) -2
3. A particle moves in a circle of radius 1m at a speed given by $v = 2t$ where v is in m/s & t in seconds, then at $t = 1$ sec.
 (A) Centripetal acceleration will be 4 m/s^2 . (B) Tangential acceleration will be 2 m/s^2 .
 (C) Tangential acceleration will be 4 m/s^2 . (D) Total acceleration will be $2\sqrt{5} \text{ m/s}^2$.
4. The second-hand of a tower clock is 1.00 m long. Which of the following is **CORRECT** statement.
 (A) Linear velocity of the tip of the hand is $\frac{10\pi}{3} \text{ cm/s}$.
 (B) Angular velocity of the hand is $\frac{\pi}{30} \text{ rad/s}$.
 (C) Average acceleration of the tip of the hand in 15 minutes is zero.
 (D) Instantaneous acceleration of the tip of the hand is $\frac{\pi^2}{9} \text{ cm/s}^2$.

Space for Rough Work

5. A uniform heavy rope of mass 8 kg and length 100 cm is being pulled as shown. Choose **CORRECT** statement(s) :



- (A) Acceleration of rope will be 5 m/s^2 .
 (B) Tension in rope at 25 cm from it's left corner will be 10 N.
 (C) Tension in rope at 75 cm from it's left corner will be 30 N.
 (D) Tension in rope at 50 cm from it's left corner will be 20 N.
6. Ideal pulley is pulled by force F upwards. If 3 kg moves with constant velocity, choose **CORRECT** statement(s) :



- (A) Tension in string is 30 N.
 (B) Acceleration of 2 kg block is 5 m/s^2 .
 (C) Force F is 60 N.
 (D) Acceleration of 3 kg block is zero.

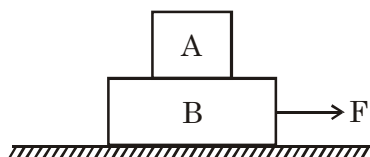
Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Paragraph for Questions 7 and 8

In a 2-block system, an external force F is applied on block B as shown in figure. Coefficient of friction between block A & block B is $\mu = 0.6$ and there is no friction between block B and ground surface. Accelerations of block A and B are a_A & a_B respectively. Friction force acting between A & B is f . Masses of blocks A & B are 2kg and 4 kg respectively.

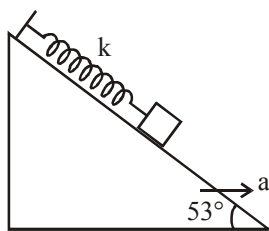


7. Mark the correct if $F = 18 \text{ N}$, then :
 (A) A & B move together (B) $a_A = 5 \text{ m/s}^2$
 (C) $a_B = 5 \text{ m/s}^2$ (D) $f = 20 \text{ N}$
8. Mark the correct if $F = 48 \text{ N}$, then :
 (A) A & B won't move together (B) $a_A = 15 \text{ m/s}^2$
 (C) $a_B = 15 \text{ m/s}^2$ (D) $f = 50 \text{ N}$

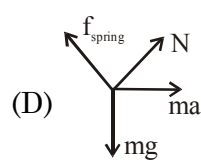
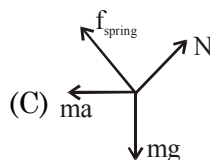
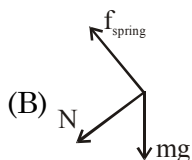
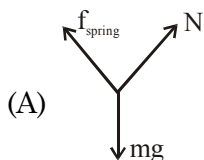
Space for Rough Work

Paragraph for Questions 9 and 10

A block of mass m connected to a spring on a wedge as shown in figure. Wedge is accelerated towards right. (All contact surfaces are smooth.)



9. Which of the following free body diagram is **CORRECT** w.r.t wedge :-



10. Which of the following statement is **CORRECT**

(A) If $a = g$ spring will be compressed.

(B) If $a = \frac{4g}{3}$ spring will be in natural length

(C) Normal reaction between block and wedge reduces as acceleration of block is increased.

(D) None

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

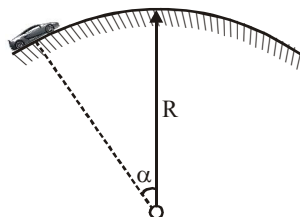
- This section contains **EIGHT** questions.
 - The answer to each question is a **NUMERICAL VALUE**.
 - For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.
- For Example :** If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

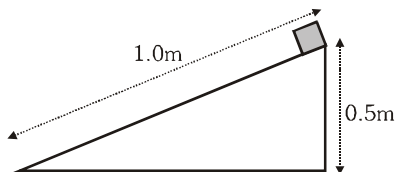
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. An escalator is used to move 5 people (60 kg each) per minute from the first floor of a department store to the second floor, 20 m above. Neglecting friction calculate the power (in kW) required.
2. The vehicle of mass $m = 100$ kg moves with speed v of a convex bridge, having a radius of curvature $R = 5$ m. What should be the speed ' v ' (in m/s) of vehicle at the instant shown, so that it leaves the surface immediately? Take $\alpha = 60^\circ$ at this instant.

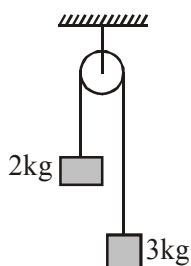


Space for Rough Work

3. A block of mass 1 kg is slipping down an incline plane of length 1 m and height 0.5 m as shown. Work done by a force is $\vec{F} \cdot \vec{d}$ where $\vec{F}$ is the force and $\vec{d}$ the displacement. Find the work done (in Joule) by weight of the block during slipping down to the end of the incline plane. (Take $g = 10 \text{ m/s}^2$)



4. In the ideal atwood machine arrangement shown, what is the change in kinetic energy (in Joule) of the system in $\frac{1}{\sqrt{2}}$ second of their motion starting from rest ?



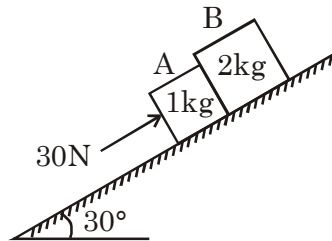
Space for Rough Work

5. A cyclist speeding at 2 m/s on a level road takes a sharp circular turn of radius 2m without reducing the speed. If minimum coefficient of static friction between tyres and road so that cyclist does not slip while taking the turn is $\frac{1}{n}$, find value of n.
6. Initially system was at rest. If acceleration (in m/s^2) of block A just after spring between block A & block B is cut is $5n$. Find value of n.

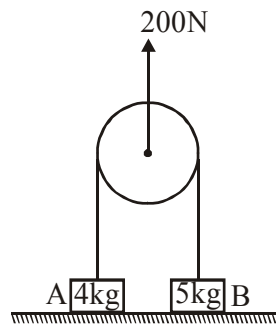


Space for Rough Work

7. If normal contact force between block A & block B is $5n$ Newton, find value of n .



8. Two blocks of masses 4 kg and 5 kg are at rest on the ground. The masses are connected by a massless string passing over a smooth and light pulley. A 200 N force is applied on the pulley. Find the difference in accelerations of blocks in m/s^2 .



Space for Rough Work

PART-2 : CHEMISTRY**SECTION-I(i) : (Maximum Marks: 24)**

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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following molecular species is/are planar and non polar :
 (A) XeF_5^- (B) XeF_2 (C) XeF_4 (D) XeF_3^+
2. If internuclear axis of $\text{O} = \text{C} = \text{O}$ molecule is z-axis, then select **CORRECT** option/s.
 (A) $\text{C} - \pi - \text{O}$ bond is formed by overlapping of $2p_x$ of carbon with $2p_x$ of oxygen atom
 (B) $\text{C} - \pi - \text{O}$ bond is formed by overlapping of $2p_y$ of carbon with $2p_y$ of oxygen atom
 (C) Nodal plane of both π -bonds do not pass through z-axis
 (D) $2p_z$ orbital of carbon participates in hybridisation

Space for Rough Work

3. Solution(s) containing 40 g NaOH is/are
- (A) 50 g of 80% (w/w) NaOH
 - (B) 50 g of 80% (w/v) NaOH [$d_{\text{soln.}} = 1.2 \text{ g/ml}$]
 - (C) 50 g of 20 M NaOH [$d_{\text{soln.}} = 1 \text{ g/ml}$]
 - (D) 50 g of 5m NaOH
4. 100 mL mixture of CO and CO₂ is mixed with 30 mL of oxygen gas and sparked in eudiometer tube. The residual gas after treatment with aq. KOH has volume of 10 mL, which remains unchanged when treated with alkaline pyrogallol. If all the volume are under same conditions point out the correct option(s) :
- (A) The volume of CO that react is 60 mL
 - (B) The volume of CO that remains unreacted is 10mL
 - (C) The volume of O₂ that remains unreacted is 10 mL
 - (D) The volume of CO₂ that gets absorbed by aq. KOH is 90 mL
-

Space for Rough Work

5. In which of the following pair of molecules/ions the underlined atoms are having same oxidation states -
- (A) $\underline{\text{Br}}\text{F}_5$, $\underline{\text{P}}\text{Cl}_6^-$ (B) $\text{H}_3\underline{\text{P}}\text{O}_2$, $\underline{\text{Cr}}\text{F}_3$
 (C) $\text{H}_2\underline{\text{S}}\text{O}_5$, $\underline{\text{Cr}}\text{O}_5$ (D) $\underline{\text{N}}_2\text{O}_3$, $\underline{\text{Cl}}\text{O}_2^-$
6. To prepare a fertiliser, an engineer dilutes a stock solution of sulphuric acid by adding 25m^3 of 7.5 M acid to enough water to make 500 m^3 of solution, which the following is/are correct about final solution-
- (A) Molarity of final solution is 0.375
 (B) Mass of sulphuric acid present in 1 litre of final solution is 36.75 gm
 (C) 500 ml of this final solution can neutralise 7.5 g of NaOH
 (D) Total moles of sulphuric acid present in final solution is 187.5

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

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Paragraph for Questions 7 and 8

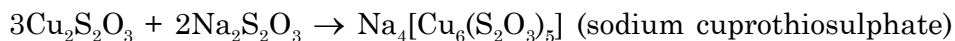
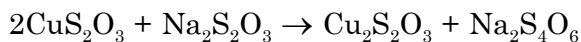
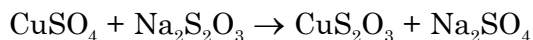
Equal number of atoms are contained in one gram-atom of each element, and that the same number of molecules is found in one gram -molecule of any compound. The terms gram atomic weight and gram molecular weight are used to refer to a fixed number (Avogadro's number 6.022×10^{23}) of particle. The term mole stands for the amount of material which contains these number of particles.

7. If 12 g of C^{12} has equal number of atoms as Avogadro's number what mass of ${}^4_2He^{2+}$ will contain same number of atoms -
 (A) 2 g (B) 4 g (C) 6 g (D) 3 g
8. What will be the number of total electrons in 24 gm ${}^{24}_{12}Mg^{2+}$
 (A) 12 (B) 10 (C) $12 N_A$ (D) $10 N_A$

Space for Rough Work

Paragraph for Questions 9 and 10

632 g of sodium thiosulphate ($\text{Na}_2\text{S}_2\text{O}_3$) reacts with copper sulphate to form cupric thiosulphate which is reduced by sodium thiosulphate to give cuprous compound which is dissolved in excess of sodium thiosulphate to form a complex compound sodium cuprothiosulphate ($\text{Na}_4[\text{Cu}_6(\text{S}_2\text{O}_3)_5]$).



In this process, 0.2 mole of sodium cuprothiosulphate is formed. (O = 16, Na = 23, S = 32)

9. Moles of sodium thiosulphate reacted and unreacted after the reaction are respectively
(A) 3 & 2 (B) 2 & 3 (C) 2.2 & 1.8 (D) 1.8 & 2.2
10. If instead of given amount of sodium thiosulphate, 2 moles of sodium thiosulphate along with 3 moles of CuSO_4 were taken initially. Then moles of sodium cuprothiosulphate formed is -
(A) 0 (B) 1 (C) 1.5 (D) 2

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

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+									
●	●	○	○	○	○	○	○	○	○
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3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	○	○	○	○	○	○	○
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	●	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
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- Find out total number of molecular species in which central atom carries only two lone pairs.
 NH_2^+ , SF_4 , ClF_2^+ , ICl_4^- , XeF_2 , XeF_5^- , CH_3^- , TeF_5^- , BrF_2^-
- Find total number of atoms which absorb energy to complete their octet by gain of electron(s).
 $\text{F}_{(g)}$, $\text{Cl}_{(g)}$, $\text{Br}_{(g)}$, $\text{I}_{(g)}$, $\text{N}_{(g)}$, $\text{C}_{(g)}$, $\text{P}_{(g)}$, $\text{S}_{(g)}$, $\text{O}_{(g)}$

Space for Rough Work

3. Find the value of $\frac{|x-y|}{4}$, where

x = atomic number of element having zero unpaired e^- in 3d series

y = atomic number of element having maximum number of unpaired e^- in 3d series

4. 12 moles of each A & B are allowed to react as given : $3A + 2B \rightarrow C + \frac{1}{2}D$. If 12 g of D is produced then the atomic mass of D is.
5. 40 g of 20% w/w NaOH solution contains 'x' g of NaOH. The value of 'x' is.

Space for Rough Work

6. If 6.81 L of CO_2 is liberated from complete decomposition of 600g CaCO_3 sample at STP then % purity of CaCO_3 sample will be.
7. Given that the four isotopes of strontium ^{84}Sr , ^{86}Sr , ^{87}Sr and ^{88}Sr having abundance 1%, 10%, 7% and 82% respectively. Atomic mass of Sr is.
8. $\text{Br}_2(l)$ reacts with $\text{Cl}_2(g)$ to form BrCl and BrCl_3 , simultaneously. How many moles of $\text{Cl}_2(g)$ reacts completely with 3 moles of $\text{Br}_2(l)$ to form BrCl and BrCl_3 in 5 : 1 molar ratio ?

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
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Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. The value of $\sum_{k=1}^{100} \sin(kx) \cos(101-k)x$ can be equal to
 (A) 50 (B) 100 (C) 25 (D) -25
2. Which of the following identities wherever defined hold(s) good ?
 (A) $(\tan \theta + \cot \theta)^2 = \sec^2 \theta \operatorname{cosec}^2 \theta$
 (B) $\tan 3\theta - \tan 2\theta - \tan \theta = \tan 3\theta \tan 2\theta \tan \theta$
 (C) $\frac{\cot^2 \theta + 1}{1 - \sin^2 \theta} = \sec^2 \theta + \operatorname{cosec}^2 \theta$
 (D) $\tan \left(\frac{\pi}{4} + \theta \right) + \tan \left(\frac{\pi}{4} - \theta \right) = 2 \operatorname{cosec} 2\theta$

Space for Rough Work

3. If $\sum_{r=1}^n r(r+2)(2r+3) = an^4 + bn^3 + cn^2 + dn + e$, then which of the following hold(s) good ?

(A) $2a + 3b - c = 4$

(B) $9b^2 + c^2 + e^2 = 149$

(C) $4a^2 + c^2 = 50$

(D) $6b + e = 20$

4. Let $a_1, a_2, a_3, \dots, a_n$ be in A.P., and $g_1, g_2, g_3, \dots, g_n$ be in G.P.

If $a_1 = 3, g_1 = 2$ and $a_9 = g_9 = 5$, then which of the following hold(s) good ?

(A) $a_5 g_{17} = 50$

(B) $a_{21} g_{25} = 250$

(C) $a_{37} g_{17} = 150$

(D) $a_{29} g_9 \neq a_5 g_{17}$

Space for Rough Work

5. In ΔABC , if $\cos A + \cos B = 4 \sin^2 \frac{C}{2}$, then which of the following hold(s) good ?

(A) $\cot \frac{A}{2} \cot \frac{B}{2} = 2$

(B) $\cot \frac{A}{2} \cot \frac{B}{2} = 3$

(C) a, c, b are in A.P.

(D) a, b, c are in G.P.

6. Consider the system of equations

$$2x + \lambda y + 6z = 8$$

$$x + 2y + \mu z = 5$$

$$x + y + 3z = 4$$

The system of equations do not have solution then :

(A) $\lambda = 2$

(B) $\lambda \neq 2$

(C) $\mu = 3$

(D) $\mu \neq 3$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Paragraph for Questions 7 and 8

Consider determinant $\begin{vmatrix} a & a^2 & 0 \\ 1 & 2a+b & (a+b)^2 \\ 0 & 1 & 2a+3b \end{vmatrix}$

On the basis of above information, answer the following questions :

7. Which of the following combination will make determinant = 0
(A) $a = b = 1$ (B) $a = 1, b = -2$ (C) $a = -2, b = 1$ (D) $a = 2, b = 2$
8. Which of the following is not a factor of determinant after expansion
(A) a (B) $a - b$ (C) $a + b$ (D) $a + 2b$

Space for Rough Work

Paragraph for Questions 9 and 10

Consider a ΔABC with usual notation where $a = 4$, $b = 8$ and $\angle C = 60^\circ$.

On the basis of above information, answer the following questions :

9. Length of side C is equal to :

(A) 4

(B) $2\sqrt{3}$

(C) $4\sqrt{3}$

(D) 10

10. Value of $\angle A$ is equal to :

(A) 60°

(B) 45°

(C) 30°

(D) 90°

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

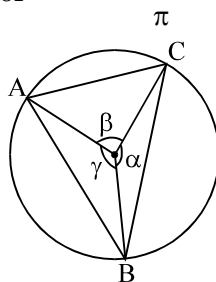
-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Value of the expression $\frac{4 + \sec 20^\circ}{\operatorname{cosec} 20^\circ}$ is
- The reciprocal of the value of the product

$$\sin \left(\frac{5\pi}{2^{11}} \right) \cos \left(\frac{5\pi}{2^{11}} \right) \cos \left(\frac{5\pi}{2^{10}} \right) \cos \left(\frac{5\pi}{2^9} \right) \dots \cos \left(\frac{5\pi}{2^3} \right) \cos \left(\frac{5\pi}{2^2} \right) \text{ equals.}$$

- A circle of area 20 is centered at the point indicated by a solid dot in the accompanying figure. Suppose that $\triangle ABC$ is inscribed in that circle and has area 8. The central angles α , β and γ are as shown. then the value of $\frac{\sin \alpha + \sin \beta + \sin \gamma}{\pi}$ is



Space for Rough Work

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4. The quadratic equation $ax^2 + bx + c = 0$ where a, b, c are the sides of $\triangle ABC$, and the equation $x^2 + \sqrt{2}x + 1 = 0$ have a common root. The measure of $\angle C$, in radian is
5. Value of $\tan \frac{\pi}{8}$ is
6. The ratio line joining points $A(8, 9)$ and $B(7, -4)$ is divided by x -axis is $\lambda : 1$ then λ is
7. Distance of point $P(3, 4)$ from x -axis is
8. Minimum integral value of ' a ' for which the equation $x^4 - 2ax^2 + x + a^2 - a = 0$ has all its roots real
-

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	24
I(ii)	Paragraph Based (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	12
II	Numerical Value Type (Up to second decimal place)	8	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator